

In-Class Task: Keys, ER Modeling & Normalization

Database Systems · Related to Laboratory Work (Relational Model & Keys)

Duration: 15 minutes

Format: Individual or pairs

Materials: Lecture notes only

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Part A: Keys (4 minutes)

An airline stores ticket data in the following relation:

```
Ticket(TicketNo, PassengerIIN, PassengerName, FlightNo, FlightDate, SeatNo, Price)
```

Business rules

- Every ticket has a unique ticket number.
- A passenger cannot hold more than one ticket on the same flight on the same date.
- A given seat on a given flight on a given date can be sold only once.
- The same flight number operates every day.

Tasks

1. List **all** candidate keys.
2. Write **three** superkeys that are **not** candidate keys, and explain why each one is not minimal.
3. Is $\{\text{PassengerIIN}, \text{FlightNo}\}$ a superkey? Justify using the business rules.
4. Which candidate key would you choose as the primary key? Give one reason.
 1. $\{\text{TicketNo}\}$, $\{\text{PassengerIIN}, \text{FlightNo}, \text{FlightDate}\}$, $\{\text{FlightNo}, \text{FlightDate}, \text{SeatNo}\}$
 2. $\{\text{TicketNo}, \text{PassengerName}\}$, $\{\text{PassengerIIN}, \text{FlightNo}, \text{FlightDate}, \text{SeatNo}\}$, $\{\text{FlightNo}, \text{FlightDate}, \text{SeatNo}, \text{Price}\}$
 3. Нет. Потому что правило "the same flight number operates every day" — то есть один и тот же FlightNo повторяется изо дня в день. Значит, пассажир вполне может иметь два билета с одинаковым $\{\text{PassengerIIN}, \text{FlightNo}\}$ — просто на разные даты (например, летает этим рейсом каждую неделю). Правило запрещает дублирование только "on the same flight on the **same date**" — а раз даты нет в наборе, уникальность не гарантирована.
 4. $\{\text{TicketNo}\}$

Part B: Functional Dependencies & 3NF (6 minutes)

A university library tracks loans in a single table:

```
Loan(ReaderID, ReaderName, Faculty, CopyID, ISBN, BookTitle, LoanDate, DueDate)
```

Business rules

- Each reader has one name and belongs to one faculty.
- The library owns several physical copies of each book. Each copy has a unique CopyID.
- Each copy belongs to exactly one ISBN, and each ISBN has one title.
- A reader may borrow the same copy multiple times, but not twice on the same day.
- The due date depends on the specific loan.

Tasks

1. Determine the primary key of `Loan`.
2. List all non-trivial functional dependencies in the form $A \rightarrow B$.

3. Give **one concrete example** each of an update, insert, and delete anomaly.
4. Identify which dependencies are **partial** and which are **transitive**.
5. Decompose `Loan` into 3NF. For each resulting table, underline the primary key and mark foreign keys.

1. `{ReaderID}, {CopyID}, {LoanDate}`
2. `ReaderID > ReaderName, Faculty; CopyID > ISBN; ISBN > BookTitle; ReaderID`
- 3.

Part C: Mini ER Diagram (5 minutes)

Scenario: Almaty City Bike Sharing

- The city has **stations**, each with a unique station code, a name, and an address (street, district).
- Each station has **docks** numbered 1, 2, 3, ... Dock 5 at one station is different from dock 5 at another.
- **Bikes** have a unique bike ID, a model, and a purchase date. The bike's age is calculated from the purchase date.
- **Users** have a unique user ID, a name, and one or more phone numbers.
- A user **rents** a bike. Each rental records the start time, end time, start station, end station, and cost.

Tasks

1. Identify all entities and state which are **strong** and which are **weak**. For each weak entity, name its owner and its partial key.
2. For each entity, list its attributes and classify each as simple, composite, multi-valued, or derived.
3. Sketch the ER diagram with cardinalities (1:1, 1:N, M:N) and participation (total/partial).
4. Decide whether *Rental* should be a relationship or an entity. Justify your choice in 1–2 sentences.

